MADAN MOHAN KUMAR

Software Engineer – Java/Spring Boot/NodeJS| Bengaluru, Karnataka
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PROFESSIONAL SUMMARY

Software Engineer with 3 years of experience delivering Java/Spring Boot, Node.js and ReactJS applications across REST APIs, JPA/Hibernate data layers, microservices and relational databases. Skilled in REST APIs, System design, application security and comfortable in SDLC, debugging, review, in Agile teams.

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Spring Security, JPA/Hibernate, Node.js, React.js, JavaScript
Databases & Messaging: PostgreSQL, PostGIS, MySQL, MongoDB, Redis, Kafka, SQL indexing and optimization
Cloud & DevOps: AWS (EC2, S3, SQS, IAM, ECS, RDS), Docker, GitHub Actions, AWS CDK
Practices: REST APIs, JWT/RBAC, Microservices, Database Transactions, Flyway Migrations, Data Structures & Algorithms

PROJECTS

- **Remote Code Execution Engine** | *Spring Boot*, *WebSocket*, *Docker*, *Redis*, Kafka, K6 (Load Tested 5k+ users), p95
  - Designed an asynchronous RCE pipeline where API instances enqueue execution jobs in Redis, **multiple workers** consume jobs, and each execution runs in an **isolated Docker** container with per-execution CPU/memory limits.
  - Implemented a **Redis-backed job queue** with a **fixed worker pool**, diagnosed and enforcing bounded execution concurrency by allowing each worker to process one job at a time.
  - Integrated Apache Kafka for execution-output event processing through a consumer group, enabling **real-time** client updates via a WebSocket gateway.
  - Implemented concurrent session and process tracking for cancellation, kill-on-rerun using runId correlation, execution timeouts, streamed output, and process/listener cleanu on **graceful shutdown**.
- **Patient Care Mesh** | Spring Boot, Kafka, Protobuf, GRPC, Docker, AWS CDK
  - Built event-driven Spring Boot **microservices**: patient-service publishes patient events to Kafka, keyed by patient ID to preserve per-patient ordering, and analytics-service consumes them. Events are defined in Protobuf, and billing is called synchronously over gRPC.
  - Verified 3-partition consumer group on 3 analytics instances: after a kill (ARN), **group rebalanced** and recovered to zero lag.
  - Routed traffic through a Spring Cloud Gateway behind an ALB, with JWT validation against a separate auth-service.
  - Deployed a CDK (Java) stack of ECS Fargate, RDS and ALB to the Floci emulator. Found stubbed MSK and Cloud Map resources in its logs and worked around them, and fixed a stale-image cache with **per-deploy tags**.
- **Tutor School** | *React.js*, *Node.js*, *PostgreSQL/PostGIS*, *AWS*, *Nginx*
  - Built secure APIs for Teacher/Parent/Admin dashboards with JWT auth and **RBAC middleware**, with no incidents in pilot.
  - Implemented geospatial teacher search using PostGIS, achieving sub-200ms latency across 2,000+ profiles.
  - Deployed to AWS **EC2 behind Nginx** with GitHub Actions CI/CD and CloudWatch monitoring.

WORK EXPERIENCE

System Engineer | Tata Consultancy Services (Johnson & Johnson engagement) | 3 YOE
Jul 2023 – Present
Engineered 20+ **role-based REST APIs** with JWT authentication and **RBAC** across 3+ internal applications serving 500 users.

- Designed a Spring Boot assessment platform (User, Test, Question Bank, Test Attempt domains) with UUID primary keys and Flyway-managed schema migrations.
- Implementing stateless JWT auth via Spring Security, with query design that avoids N+1 issues in the JPA/Hibernate layer.
- Designed and optimized SQL and NoSQL schemas in PostgreSQL and MongoDB, improving indexing and query performance and cutting average query **response time by 20%** on large datasets.
- Built responsive React.js front ends with REST API integration, **reducing duplicate UI code** by 30% across 5 modules.
- Participated in Agile/Scrum teams, managing **sprint** tasks in Jira, conducting code reviews, debugging, deployment, and production support, while delivering 4+ product features.

EDUCATION

MCA – SASTRA Deemed University, Chennai | CGPA: 8.3 / 10
BCA – Amity University, Patna | CGPA: 8.5 / 10

CERTIFICATION - AWS Certified Developer – Associate (DVA-C02) | Valid through Dec 2028